Chapter 7
Building ArcGIS Desktop Add-Ins

An ArcGIS desktop add-in is a software program that runs in ArcGIS environment. An ArcGIS add-in can include multiple add-in components such as buttons, tools, combo boxes, tool palettes, toolbars, dockable windows, etc. This chapter uses an example to demonstrate the entire process of creating new ArcGIS desktop add-in projects. The example will show step-by-step a process that uses the ArcGIS add-in wizard to create an add-in button, a tool, a combo box, and a toolbar. It will also show how to implement functionality of these add-in elements by using the ArcGIS Snippet Finder. An ArcGIS add-in project uses an Extensible Markup Language (XML) document to manage configuration parameters of all add-in elements. To help understand the add-in configuration document, this chapter will provide a brief introduction to XML and then explain each XML element in the configuration file. Other related techniques such as manipulating add-in elements and toolbars as well as using the ArcGIS Desktop Add-in Manager to installing/removing add-ins will also be introduced in this chapter.

1. ArcGIS Desktop Add-In

To use a Windows software program, you must install it on your computer. Installing software on a PC is usually accomplished by running an installation program which creates folders and copies files onto your computer. An installation process also registers components (some special files) in your Windows registry. The Windows registry is a database that stores configuration settings as well as component registration information. Any COM component or .NET assembly contained in a software program must be registered in Windows so that the operating system knows it and can use it when needed. When you uninstall an application, an uninstalling program is executed to delete files and unregister unused components from the Windows registry. With ArcGIS 9 and 8, any COM based additional tool is required to be registered in Windows by the user. Manually registering and unregistering components are tedious and prone to errors.

An ArcGIS desktop add-in is made up of a number of files including .NET assemblies and an Extensible Markup Language (XML) document. All these files are compressed into one zip archive marked with file extension ".esriAddin". To use a desktop add-in, you do not need to run an installation program or register assemblies in Windows like other Windows applications. If you have an ArcGIS Desktop add-in file, all you need to do to install it in ArcGIS is to double click on the file and follow instructions to complete. This makes ArcGIS add-ins easy to use and share.

Once an add-in is installed in ArcGIS, you can find it in the add-in manager of ArcGIS by clicking menus: Customize >> Add-In Manager. Figure 1 shows an add-in example that performs quadrat analysis for point patterns. This add-in was created as an ArcMap extension with a tool bar named "Point Pattern Analyst". The toolbar contains a combo box that lists of all point feature classes loaded into the map, an extent tool that can be used to draw a rectangle in the map to define an analysis extent, and a button to show/hide a dockable window. The dockable window displays a GUI that lets the user to specify an analysis extent, to edit the analysis extent, to specify a quadrat size, and to start computing. Result of computation is displayed in a text box in the dockable window (Figure 1).

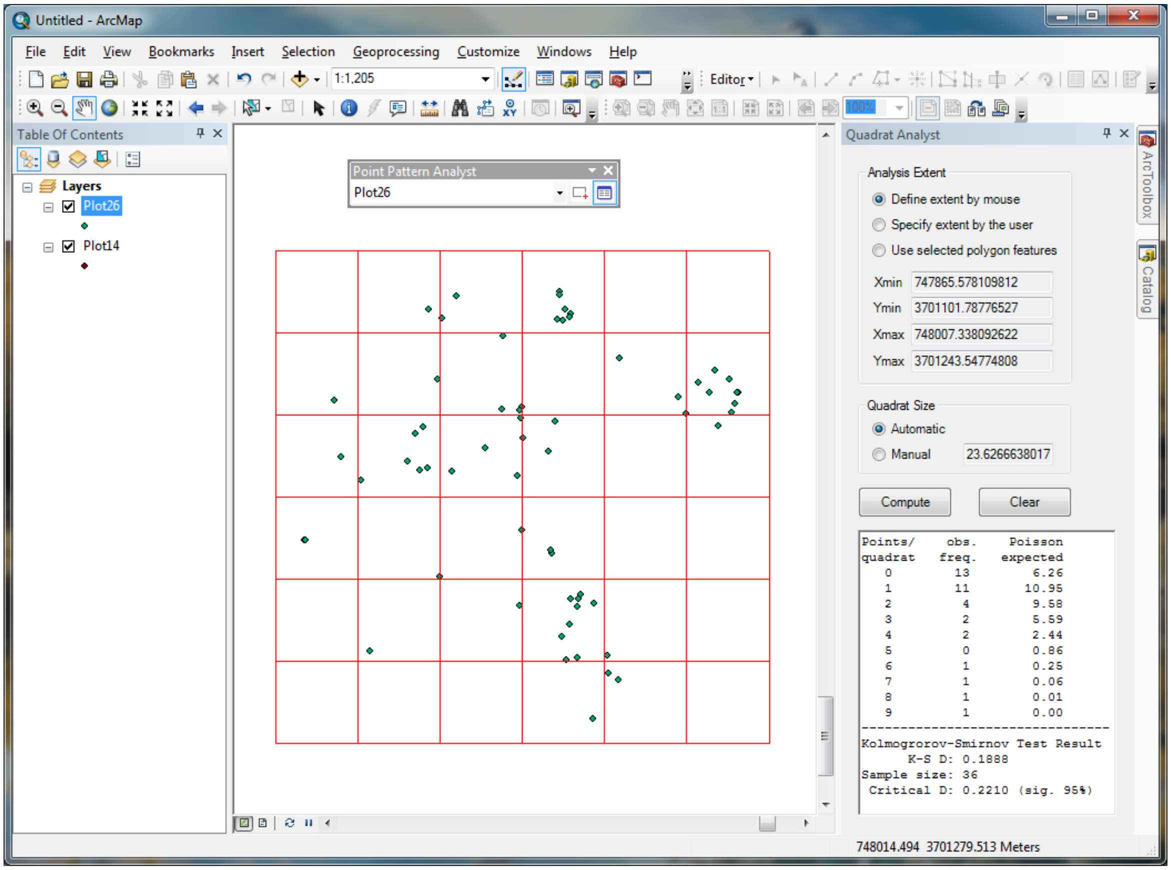

Figure 1 An example of an ArcGIS desktop add-in

ArcGIS 10 desktop applications, such as ArcMap and ArcCatalog, support a number of add-in controls including buttons and tools, combo boxes, menus and context menus, multi-items, toolbars, tool palettes, dockable windows, application extensions, and editor extensions. To get started, this chapter only introduces the following four frequently used add-in controls. More add-in types will be introduced in other chapters when they are necessary.

- **Buttons** and **tools**: Buttons and tools are simple controls that can appear on toolbars. Like the windows button control, an add-in button is a control that performs tasks when clicked. In ArcMap, for example, the Save, Print, and Full View are buttons. A tool is a toggle button that is selected when clicked. Nothing else happens until the user uses the mouse to click or drag on the map. For example, the zoom in, zoom out and pan toggle buttons are tools
- **Combo boxes**: A combo box provides a drop-down list containing items for the user to select. An add-in combo box can be configured to allow users to enter values like using a text box.
- **Toolbars**: A toolbar can hold buttons, tools, menus, tool palettes, and combo boxes, all of which can be either built-in or created in the current add-in project.

2. Creating a new ArcMap add-in project

The following will guide you through the process to create an ArcMap add-in project and then create an add-in button and a tool in the project. The button will be used to zoom to the active layer in the map when it is clicked once. When the tool is selected, the user can click on the map to draw a line (and double-click to finish the line). First, let's create a new ArcMap add-in project in VB.

1) Start Visual Basic 2010.
2) If the Start Page displays, select the "New Project" item on the page. Otherwise, select in the main menu File >> New Project. The New Project dialog box opens.
3) In the Installed Templates panel, expand the "Visual Basic" node, then the "ArcGIS" node, and click "Desktop Add-Ins" (Figure 2).

> *If there are no templates in the Templates pane associated with the Desktop Add-Ins node, verify that your target .NET Framework is set to .NET Framework 3.5 in the drop-down menu in the upper right corner of the New Project dialog box.*

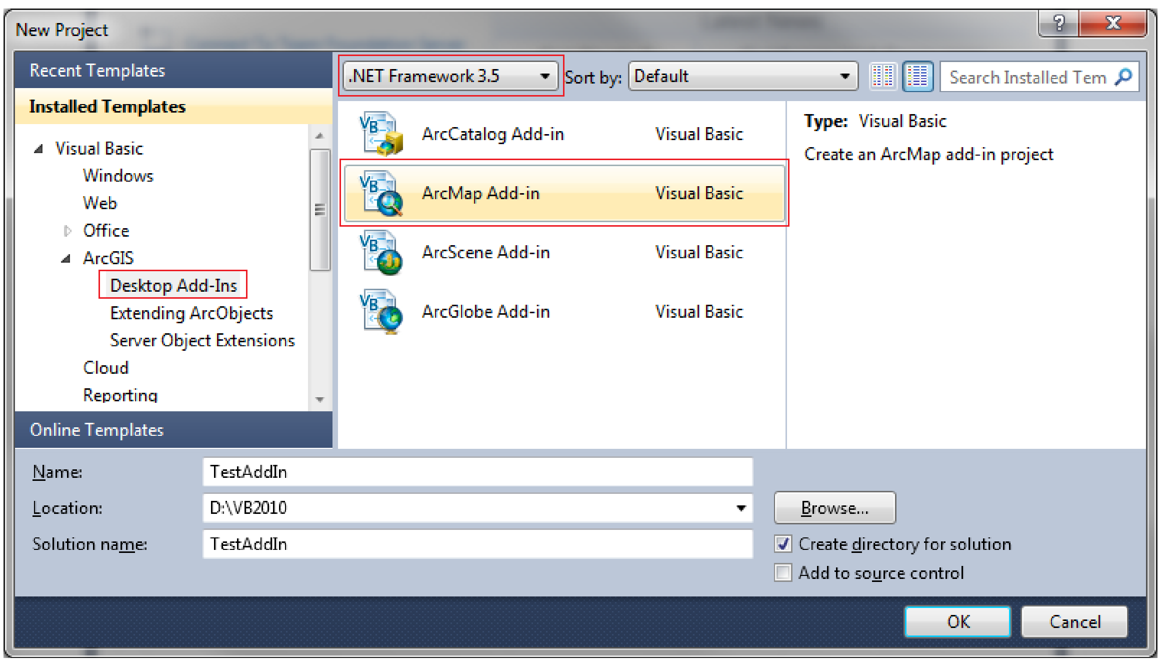

Figure 2 ArcGIS desktop add-in templates

4) Select the "ArcMap Add-in" template in the central panel (be careful, ArcCatalog add-in template is often accidently selected by students since it is on the top). This panel also contains other templates include ArcCatalog, ArcScene, and ArcGlobe Add-ins.
5) Name the project "TestAddIn" and browse to a folder, e.g. D:\VB2010, where you want to save the project (Figure 2).
6) Click OK to close the dialog box and open the ArcGIS Add-Ins Wizard.

3. Creating a button and a tool

Once a new ArcGIS add-in project is started, the ArcGIS Add-Ins Wizard opens to let you create necessary add-in elements (controls). Follow the wizard to create a button and a tool.

1) Fill in basic information about the new add-in on the Welcome page of the wizard (Figure 3): add-in name, company/publisher, author, and description. You may specify a 32x32 pixel image as an icon for the add-in project. The ArcGIS Developer Kit comes with more than 4000 standard ArcGIS images for use. These images are included in a zip archive file “Icons.zip” on the CD-ROM. The information you provide on the Welcome page will be displayed in the Add-In Manager dialog. Click the “Next” button to continue.

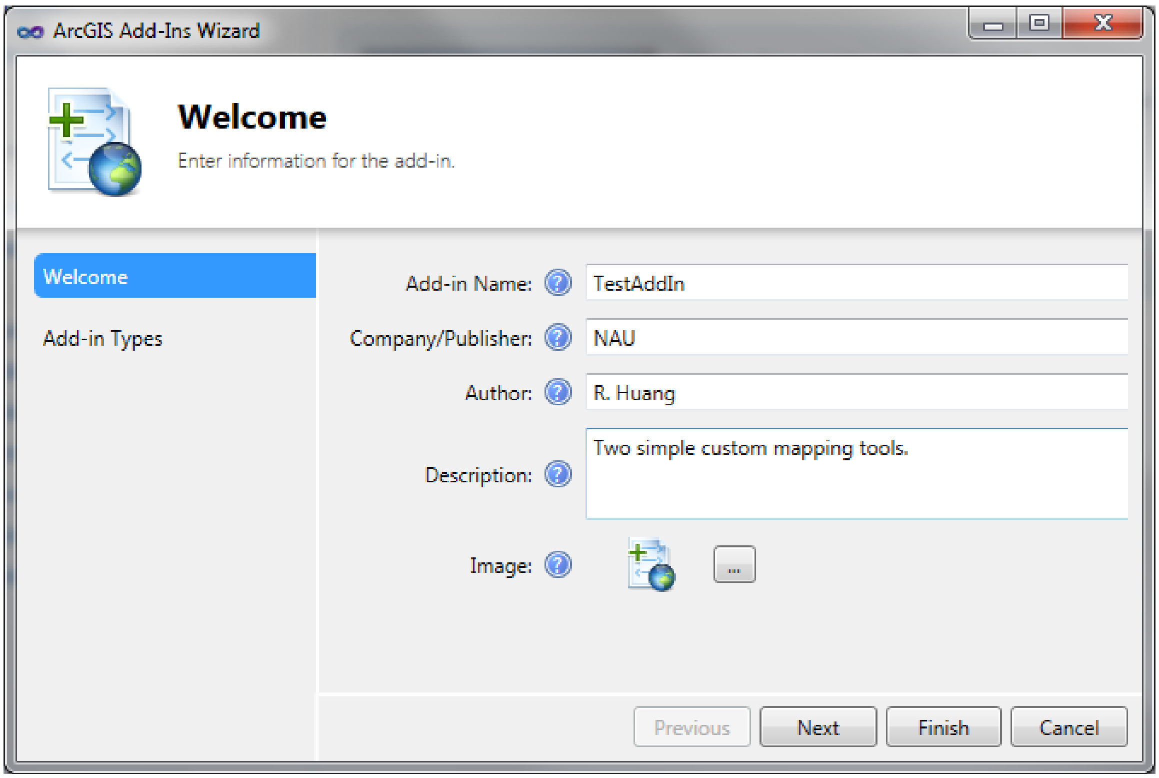

Figure 3 ArcGIS add-in wizard welcome page

2) On the “Available Add-in Components” page you can find a list of add-in component types in a panel on the left. Please check the “Button” check box. You will find the declarative properties of the button including button name, caption, image, and tooltip become active.
3) Fill in the button information by following the example of Figure 4. Choose a 16 x 16 pixel icon for the add-in button. If you are not sure about the purpose of an entry, you can find help tips by hovering your mouse over the appropriate help icon. Once a “ZoomToLayerButton” button is created, a class with the same name will be created by the wizard. The ZoomToLayerButton class is where you will write code to implement the zoom task.

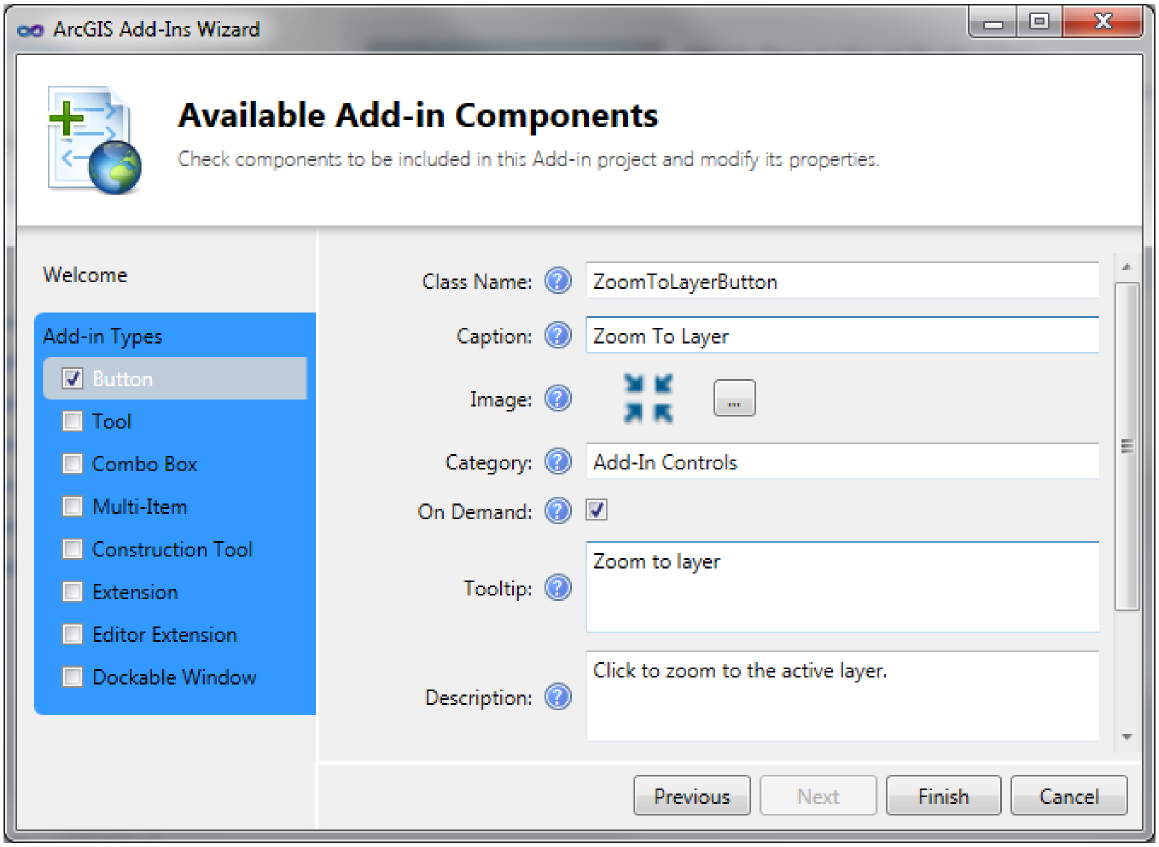

Figure 4 ArcGIS add-in wizard button configuration page

5) After creating the button, check the “Tool” check box to create a custom tool in the add-in project. Enter “AddGraphicsTool” as the class name and fill in other declarative properties of the tool as shown in Figure 5. You can also select a 16 x 16 pixel icon for the tool. Since we do not have a cursor file, you do not have to specify a cursor to it yet. A blank class named AddGraphicsTools will be created by the wizard, and you will write code in the class to implement functions (Figure 5).

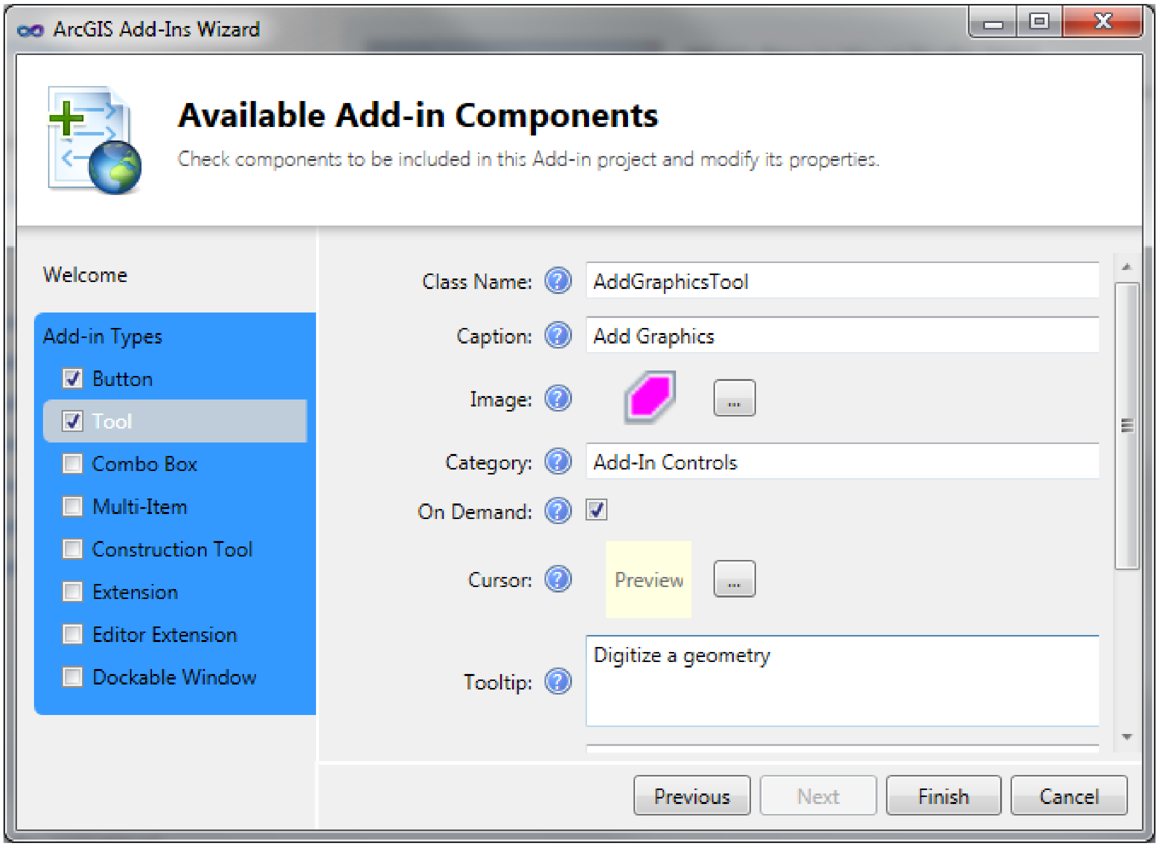

Figure 5 ArcGIS add-in wizard tool configuration page

7) Click the “Finish” button to finish the wizard. The screenshot of Figure 6 shows the files created by the wizard as seen in the Solution Explorer.

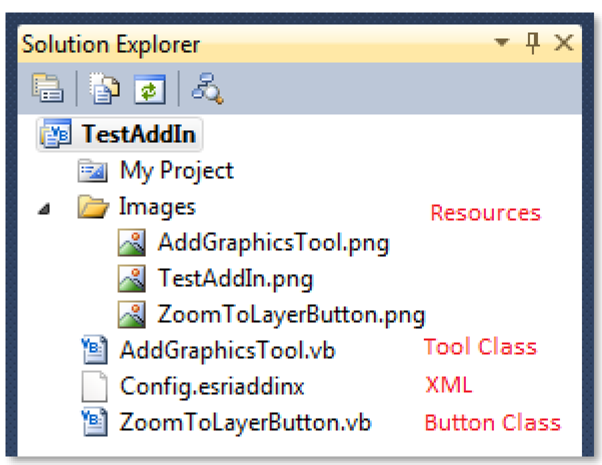

Figure 6 Fires generated by the ArcGIS desktop add-in wizard

In the Solution Explorer, you can find that the images and cursors you specify to the button and tool are copied to the Images folder. Two class modules (.vb files) are created for the button and tool implementation. File “Config.esriadddinx” is an XML document containing XML elements that define the declarative properties of the button and tool. The following section introduces XML and explains the contents of the XML in detail.

4. Understanding the XML

The XML document (file “Config.esriaddinx”) generated by the wizard describes the add-in project and its custom components. Before explaining this document, it is necessary to introduce the Extensible Markup Language (XML). XML is not a programming language; instead it is a text document used to encode information or data in custom-defined structures. Similar to the popular Hypertext Markup Language (html) used to encode web pages, XML also uses tags to encode data. Unlike html which only uses a limited number of fixed tags, XML lets the user to define tags. In XML, an element, which is a block of information, must have a beginning tag and a closing tag. For example,

```
<note>Welcome to GIS Programming</note>
```

is a user defined element named note. The data of the element, “Welcome to GIS Programming”, is enclosed in a pair of user defined tags `<note> </note>`.

An element may have attributes which are defined in the beginning tag after the tag name. An attribute must be assigned a value, and the value must be quoted by a pair of quote signs no matter it is a number or a string. For example, we can create an attribute named *id* for the note element:

```
<note id="1 ">Welcome to GIS Programming</note>
```

Some XML element may only have attributes without enclosed data. In this case, the element does not have to have a closing tag, but the element must be closed by a slash symbol (/) in the end of the beginning tag. For example,

```
<button id="1001" name="zoom_button" image="zoomicon.png" />
```

In XML, one element can be nested in another to make a hierarchically structured document. Theoretically, there is no limit in number of levels in element nesting. Therefore XML is perfect for encoding information with hierarchical structure. For example,

```
<course prefix="GSP" number="435" name="Programming for GIS">
      <note id="1">
            <from>Instructor</from>
            <to>All students</to>
            <date>January 12, 2014</date>
            Welcome to GIS Programming    ← text data of the note element
      </note>
      <note id="2">
            ...
      </note>
</course>
```

The xml document above has three levels of elements: the first level is the *course* element. This element has three attributes: *prefix*, *number*, and *name*. The second-level consists of two *note* elements each having an id attribute. The third level contains one piece of text data (“Welcome to GIS Programming”) and three elements, *from*, *to*, *date*, each containing a piece of text data.

An ArcGIS add-in project uses an XML document to maintain add-in configurations and add-in component definitions. The structure and contents of the Config.esriaddinx XML document are as follows.

1) The top level of XML element is *ESRI.Configuration*. This element has two attributes *xmlns* and *xmlns:xsi*. This top-level element contains many secondary and lower level elements.
2) Inside the *ESRI.Configuration* element, a number of metadata elements including *name*, *AddinID*, *description*, *version*, *image*, *author*, *company*, and *date* are used to describe the add-in project (Figure 7). The ArcGIS Add-in Manager relies on these elements to provide a description of the installed add-ins. The *AddInID* element uniquely identifies the add-in. To guarantee uniqueness, this is typically a globally unique identifier (GUID). The Add-in installation utility relies on the *AddInID* and *Version* elements.
3) The *Targets* element specifies the most compatible version and product of ArcGIS for the add-in. The Add-in installation utility will install the add-in only to the product and version specified in the *Targets* element. The “TestAddIn” add-in project is created for ArcGIS “Desktop” version “10.2” (Figure 7).

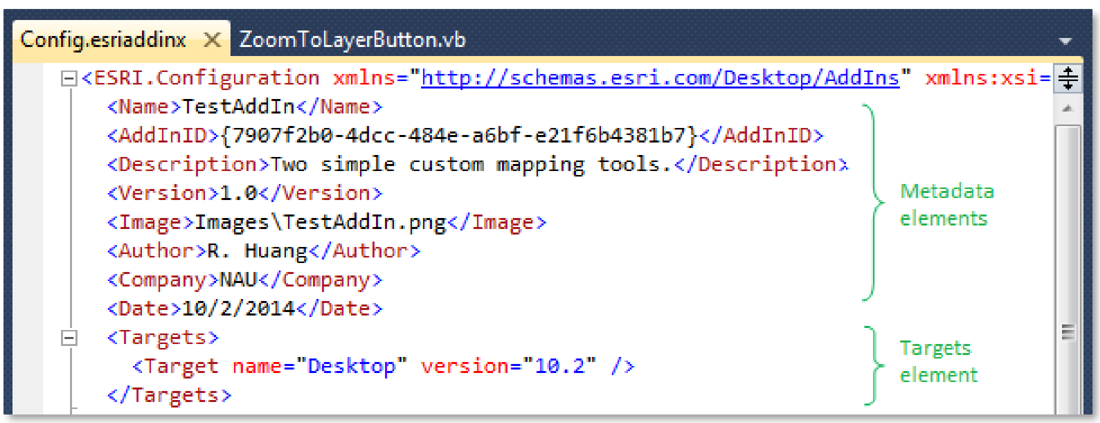

Figure 7 The metadata element section of the XML configuration document

3) The *AddIn* element (Figure 8) is the parent element that includes all the add-in components (including controls). The *language* attribute on the AddIn element specifies the programming environment of the add-in. The supported values are “CLR”, which indicates a .NET managed add-in, and “Java”, which indicates a Java add-in. The *library* attribute indicates to the add-in framework the related assembly to load. Although add-in files can contain multiple assemblies, most rely on a single assembly. The *namespace* attribute is the default namespace for all classes in the assembly. The automatically generated XML puts all attributes and their values in one long line. The long lines of elements are broken into multiple lines in Figure 8 for visual clarity. Breaking a line in an xml document at spaces between attributes does not affect validity of the document.

```xml
<AddIn language="CLR"
       library="TestAddIn.dll"
       namespace="TestAddIn">
  <ArcMap>
    <Commands>
      <Button id="NAU_TestAddIn_ZoomToLayerButton"
              class="ZoomToLayerButton"
              message="Click to zoom to the active layer."
              caption="Zoom To Layer"
              tip="Zoom to layer"
              category="Add-In Controls"
              image="Images\ZoomToLayerButton.png" />
      <Tool id="NAU_TestAddIn_AddGraphicsTool"
            class="AddGraphicsTool"
            message="Click on the map to create graphics."
            caption="Add Graphics"
            tip="Digitize a geometry"
            category="Add-In Controls"
            image="Images\AddGraphicsTool.png" />
    </Commands>
  </ArcMap>
</AddIn>
```

The beginning tag of the AddIn element with three attributes

A button element with 7 attributes

A tool element with 7 attributes

This is the closing tag of the AddIn element

Figure 8 The *AddIn* element and its child elements in the XML document

4) The *ArcMap* element denotes that the specified add-in components are for the ArcMap application. You can also create add-in custom components for ArcCatalog, ArcScene, and ArcGlobe.
5) The *Commands* element defines all customer add-in components including button, tool, combo box, etc. Any add-in component you create in a project using the add-in wizard should appear as an element of the Commands XML element.
6) The *Button* element denotes the custom *ZoomToLayer* button. The *class* attribute specifies the corresponding class in which the run time behavior of the button must be described. The *id* attribute is the unique ID of the button in the project. This element does not have included data.
7) The *Tool* element denotes the custom AddGraphics tool that you created through the wizard. The *class* attribute specifies the corresponding class that defines the run time behavior of the tool. The *id* attribute uniquely identifies the custom tool in the project. The tool is identified by the unique id when included in other components such as a toolbar or menu. You can find more information about the attributes of the Tool element by hovering your mouse over the attribute or refer to the add-in XML reference document.

5. Implementing functionality of the add-in button

Please perform the following steps to add zoom-to-layer functionality to the add-in button. Because ArcObjects is not introduced yet, the code will not be explained in this chapter. Please follow instructions very carefully to avoid making any mistake.

1) Double click item ZoomToLayerButton.vb in the Solution Explorer to open the ZoomToLayerButton class module. You will find that the custom button inherits from a class named ESRI.ArcGIS.Desktop.AddIns.Button.

2) Use the ArcGIS Snippet Finder tool to implement functionality of the zoom-to-layer button. The ArcGIS Snippet Finder tool provides code snippets for common workflows. You can insert the code in your classes as needed. Perform the following steps to implement the zoom-to-layer functionality:

   a. Right-click at a desired insertion point in the Visual Studio code editor window. The insertion point must be inside the ZoomToLayerButton class, but outside any existing procedure. It is where you normally create a new sub procedure or function. For example, you may add a couple of blank lines above the "End Class" statement and insert new code there.
   b. Select "ArcGIS Snippet Finder" from the context menu after right clicking at an insertion point. The ArcGIS Snippet Finder dialog box opens (Figure 9).

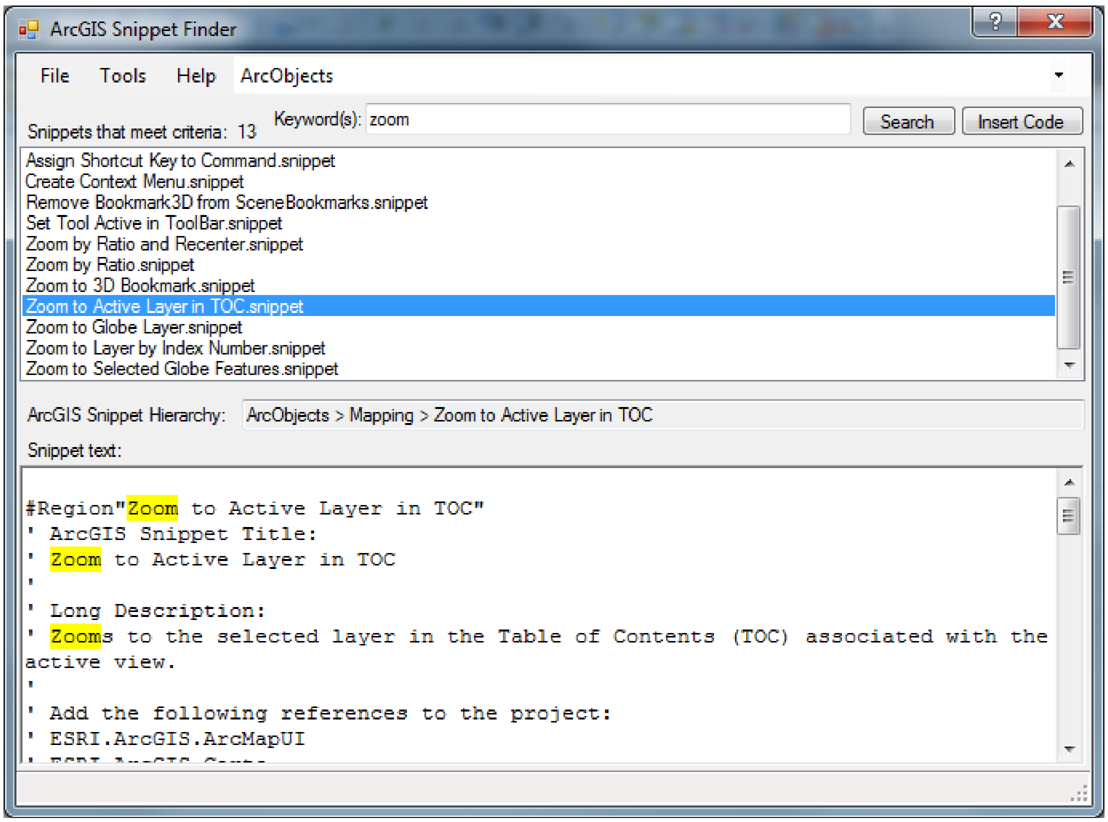

Figure 9 ArcGIS Snippet Finder

   c. Type "zoom" in the Keyword(s) text field to narrow your search of ArcGIS snippets and click the "Search" button.
   d. Select the "Zoom to Active Layer in TOC.snippet", and click the "Insert Code" button to embed the snippet into the ZoomToLayerButton class (Figure 9).

Once inserted, the snippet is a region containing a sub procedure ZoomToActiveLayerInTOC(). You will call this sub procedure in the button click event to carry out the desired task.

3) Invoke the ZoomToActiveLayerInTOC() procedure in the OnClick() event procedure by following example in Figure 10. Please carefully enter the code shown in the red box in Figure 10 into your program.

```
Protected Overrides Sub OnClick()
    '
    '  TODO: Sample code showing how to access button host
    '
    ZoomToActiveLayerInTOC(TryCast(My.ArcMap.Application.Document, IMxDocument))
    My.ArcMap.Application.CurrentTool = Nothing
End Sub
```

Figure 10 The OnClick event procedure of the zoom button

After entering the code that calls the ZoomToActiveLayerInTOC() function, you can click the run button to test the new add-in. When the program runs, ArcMap opens. Click the ArcMap menu Customize >> Add-In Manager. You will find that the new Add-in (Simple Mapping Tools) is automatically installed in ArcMap (Figure 11).

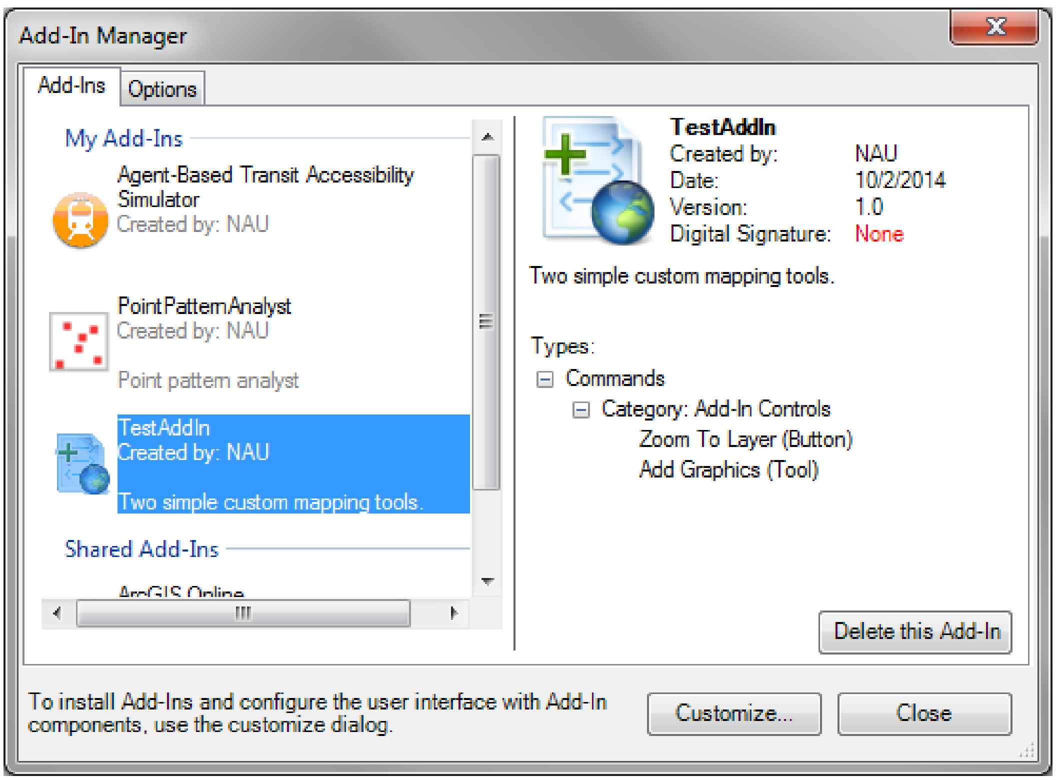

Figure 11 The ArcGIS Add-In Manager

Click the "Customize" button in the Add-In Manager window, then in the Customize dialog switch to the "Commands" tab page and select the "Add-In controls" category. You will find the "Zoom To Layer" button and "Add Graphics Tool" appearing in the commands box. Drag these two controls into any existing toolbar in ArcMap and then close the Customize dialog (Figure 12).

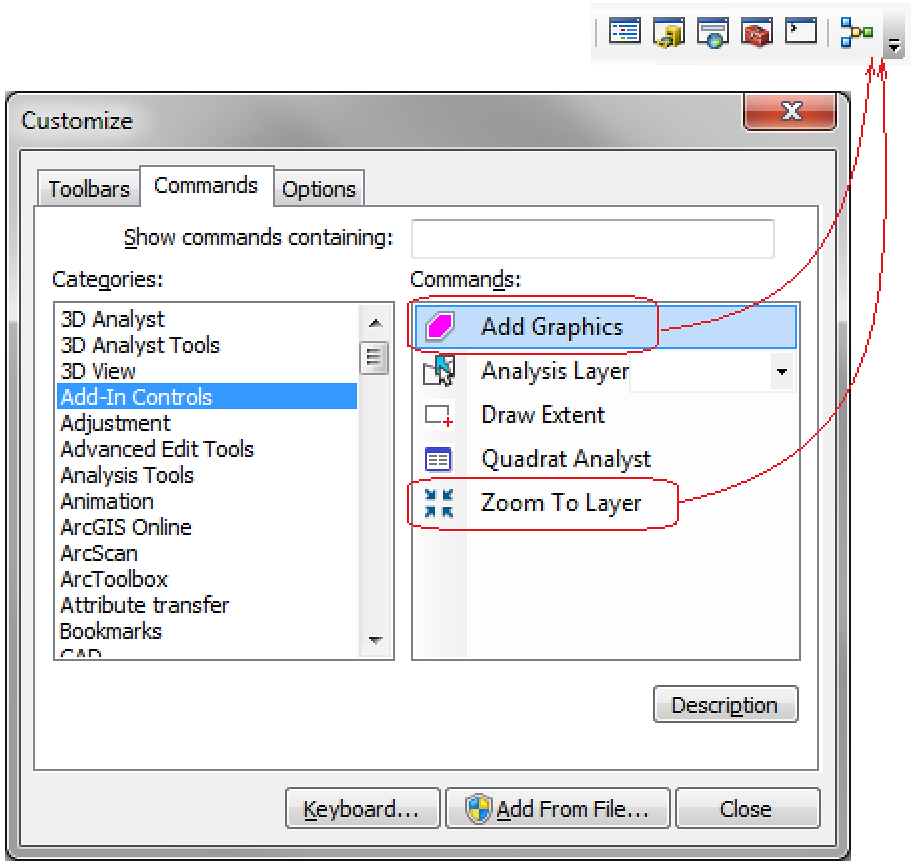

Figure 12 Adding new add-in components into ArcGIS toolbars

You can now add some map layers (better with different extents) into ArcMap and test the zoom-to-layer button. The Add graphics tool is not functioning yet since it is not implemented. After successfully testing this zoom button, you can close ArcMap to stop the program execution so as to work on the add graphics tool.

6. Implementing functionality of the add-in tool

To implement the AddGraphics custom tool, we will use the ArcGIS Snippet Finder tool again. Please perform the following steps:

1) Double click the “AddGraphicsTool.vb” file in the Solution Explorer to open the AddGraphicsTool class module. The wizard generates some plumbing code for you, but you have to create a OnMouseDown() event procedure for the AddGraphics tool.
2) Before you create the OnMouseDown() event procedure, you can use the ArcGIS Snippet Finder to add the AddGraphicToMap() function into the AddGraphicsTool class:

    a. Right-click at a desired insertion point in the AddGraphicsTool class. You are suggested to add a couple of blank lines above the “End Class” statement and insert the code there.

b. Select ArcGIS Snippet Finder from the context menu after right clicking at the insertion point. The ArcGIS Snippet Finder dialog box opens.
c. Type “add graphic” in the Keyword(s) text field and click “Search”.
d. Select the “Add Graphic to Map.snippet” and click “Insert Code”.

3) Use the snippet finder again to add polyline graphics functionality to the tool class as shown in the following steps:
   a. Type “get polyline” in the Keyword(s) text box and click the “Search” button.
   b. Select the “Get Polyline From Mouse Clicks snippet” and click “Insert Code”. This snippet will be inserted as a function.

When you program with VB 2010 in previous chapters, you can bring up a control’s event procedure by selecting in the event drop-down list in the code editor. To program for an ArcGIS add-in UI elements (a tool in this case), however, you have to manually create some event procedure blocks because an add-in element is not identical to a Windows control. In this example, you have to manually create a blank OnMouseDown() event procedure with the assistance of IntelliSense, and then add code into it to implement tasks. To do so,

4) Typing the following code in the code editor and hit the Enter key.

```
Protected Overrides Sub OnMouseDown(arg As ESRI.ArcGIS.Desktop.AddIns.Tool.MouseEventArgs)
```

As you type the code, the IntelliSense of VB keeps providing suggestions. When you see the above line highlighted in the IntelliSense, you can hit the Enter key to accept the suggestion without having to type the full line. A procedure closing statement “End Sub” will also be added by the IntelliSense once you push the Enter key.

5) Add code in the OnMouseDown() event procedure of the tool to implement the creating graphic functionality as the following.

```
Protected Overrides Sub OnMouseDown(arg As ESRI.ArcGIS.Desktop.AddIns.Tool.MouseEventArgs)
    'Get the active view from the ArcMap static class.
    Dim activeView As IActiveView = My.ArcMap.Document.ActiveView
    'If it's a polyline object, get from the user's mouse clicks.
    Dim polyline As IPolyline = GetPolylineFromMouseClicks(activeView)
    'Make a color to draw the polyline.
    Dim rgbColor As IRgbColor = New RgbColorClass()
    rgbColor.Red = 255
    'Add the user's drawn graphics as persistent on the map.
    AddGraphicToMap(activeView.FocusMap, polyline, rgbColor, rgbColor)
    'Best practice: Redraw only the portion of the active view that contains graphics.
    activeView.PartialRefresh(esriViewDrawPhase.esriViewGraphics, Nothing, Nothing)

    MyBase.OnMouseDown(arg)    ' do not touch this line
End Sub
```

Now you can run the program to test the add graphics tool. Also, please do not be too much concerned about how the code works at this point. You will understand it when you learn ArcObjects programming.

7. Creating a custom combo box

A combo box contains a drop-down list and an editable field (text box). The user can select a value from the drop-down list, which appears at the user's request. If you make the combo box editable, the combo box will include an editable field into which the user can enter a value. An ArcGIS add-in combo box is similar but not identical to a Windows combo box.

This section describes a procedure to create a combo box that adds ArcGIS Online layer data to your current document. A combo box can be included in an existing ArcGIS add-in project using the installed templates. To add a combo box element to your add-in project, perform the following steps:

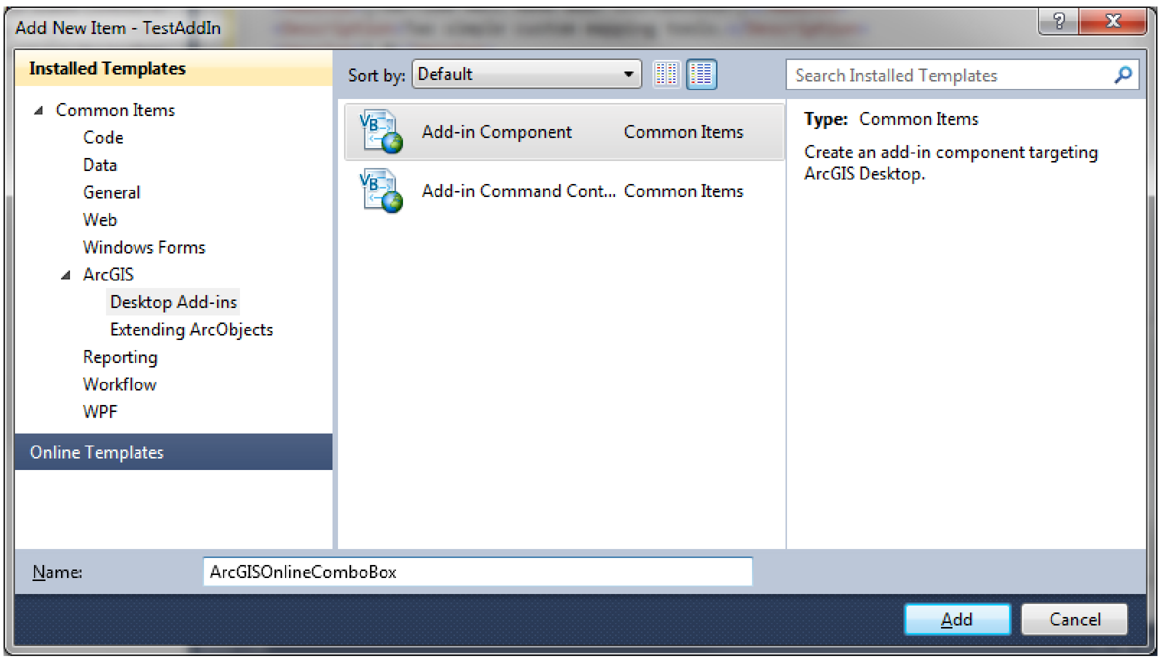

Figure 13 Adding new item dialog

1) Open the TestAddIn solution that you created previously if it's not open.
2) In the Solution Explorer, right-click on the project name "TestAddIn", then select Add, and click New Item. The Add New Item dialog box opens as shown in Figure 13.
3) Expand the Common Items node, then the "ArcGIS" node, and select "Desktop Add-ins".
4) Select "Add-in component" in the templates pane, enter a file name "ArcGISOnlineComboBox", and click the "Add" button. The ArcGIS Add-Ins Wizard opens.
5) In the "Add-in Components" section, select "Combo Box", and set values to properties of the combo box as shown in Figure 14. Please pay attention to class name, caption, tooltip and description. Two interesting properties of the add-in combo box are the "Size String" and "Item Size String". Instead of using numeric values to define the width and maximum allowed characters for this add-in element, an ArcGIS add-in project uses a number of letters ("W") to set these two properties.

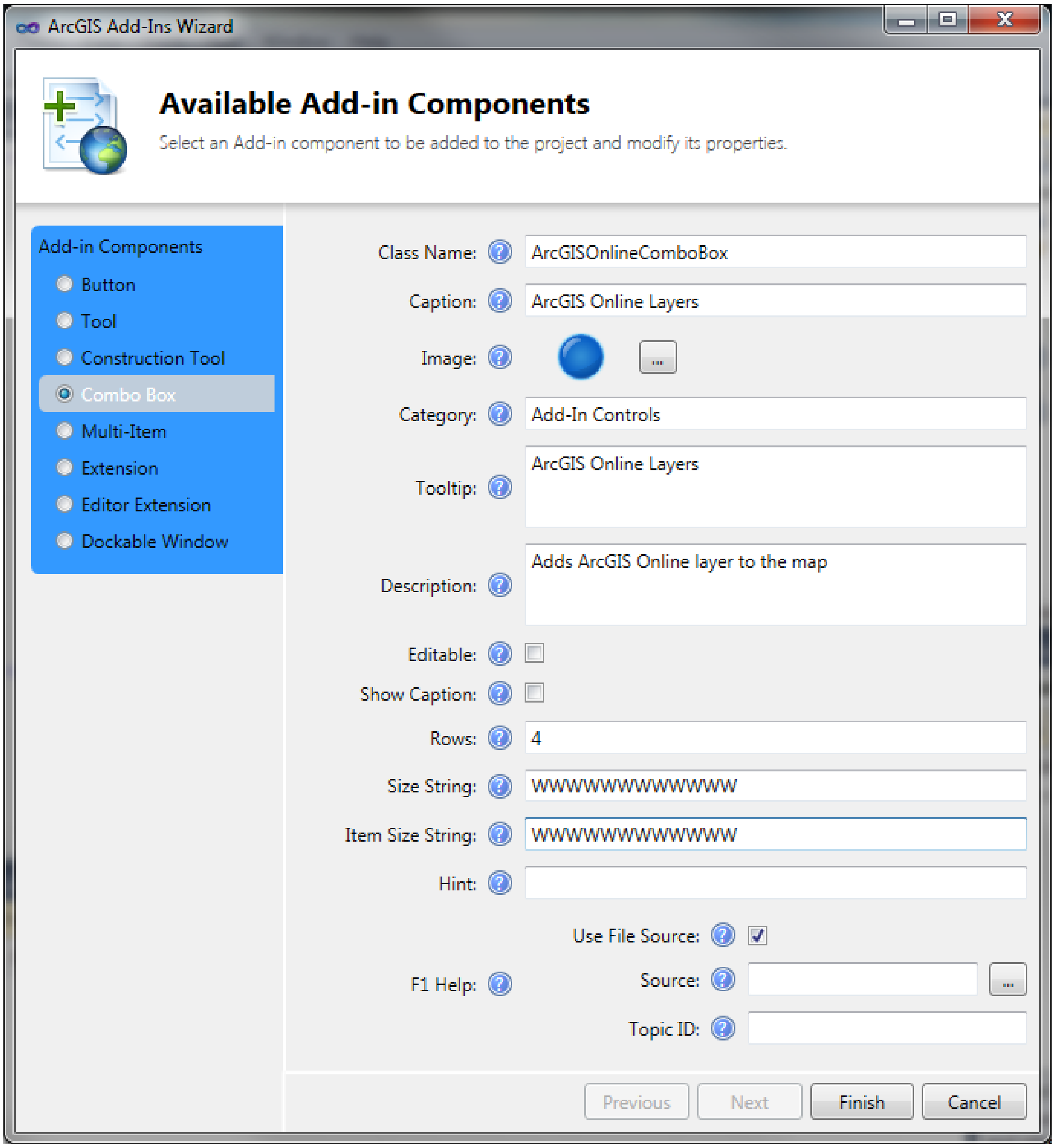

Figure 14 ArcGIS add-in wizard combo box page

6) Click Finish. The XML elements specific to the combo box are added to the project configuration file (Config.esriaddinx), the images specified in the wizard are copied to the image folder, and the .NET managed class is created in the project with some template code.
7) To add functionality to the custom combo box, copy and paste the following code into the “ArcGISOnlineCombox” class. Make sure that there is only one “New()” sub procedure.

```
Public Sub New()
    Dim serviceLinks As String() = {"Add a World Map Service", _
                                    "World_Topo_Map", _
                                    "World_Terrain_Base"}
    For Each currLink As String In serviceLinks
        Add(currLink)
    Next
```

```
    End Sub

    Protected Overloads Overrides Sub OnSelChange(ByVal cookie As Integer)
        Dim selectedURL As String = Nothing
        ' The value property provides the selected item.
        Select Case Value
            Case "World_Topo_Map"
                selectedURL = "http://services.arcgisonline.com/ArcGIS/rest/services/" & _
                              "World_Topo_Map/MapServer?f=lyr&v=9.2"
            Case "World_Terrain_Base"
                selectedURL = "http://services.arcgisonline.com/ArcGIS/rest/services/" & _
                              "World_Terrain_Base/MapServer?f=lyr&v=9.2"
        End Select
        If selectedURL IsNot Nothing Then
            System.Diagnostics.Process.Start(selectedURL)
        End If
        MyBase.OnSelChange(cookie)
    End Sub
```

8) To test the combo box, run the program and wait ArcMap to open. Then
    a. Select ArcMap menu Customize >> Customize Mode… to open the Customize dialog.
    b. In the Customize dialog, switch to the Commands tab page, select “Add-In Controls” in the Categories list, and then drag the “ArcGIS Online Combo Box” add-in into any existing toolbar.
    c. Select a map from the new combo box to display. If you see the selected map layer in a pop-up “View Downloads” window, click the “Open” button to proceed. If may a moment for Windows to download data.

8. Creating a custom toolbar

The three add-in controls you created in previous sections are currently hosted by existing toolbars of ArcMap. In an ArcGIS Desktop add-in project, you can create your own toolbars to hold controls and commands including those developed in the current project or other add-ins, and built-in commands. Perform the following steps to create a custom toolbar to hold the three add-in components you create in the TestAddIn project.

1) In the Solution Explorer of the TestAddIn project, right-click the solution name, select “New Item”, and click “Add”. The Add New Item dialog box opens.
2) In the Installed Templates panel, expand the “Common Items” node, then the “ArcGIS” node, and select “Desktop Add-ins”.
3) Select “Add-in component Container” from the central pane and click the “Add” button. You do not need to provide a name when you create a toolbar. The ArcGIS Add-Ins Wizard opens.
4) Select “Toolbar” under the Add-in Command Bars.
5) You may modify the toolbar caption in the Caption field. For example, you may use “TestAddIn Toolbar”
6) Click “Add Item…” in the Items section to add elements into the toolbar. A drop-down box appears with a list of add-in components you have created in the project (Figure 15):

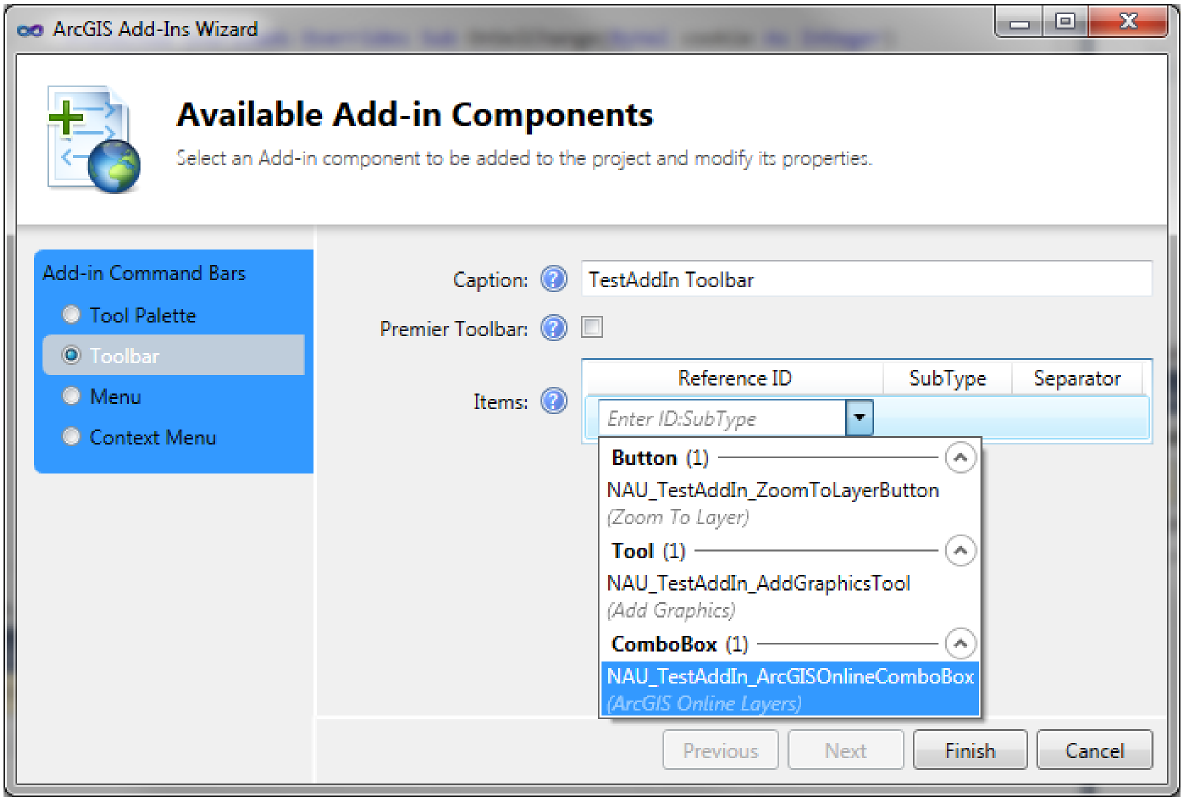

Figure 15 ArcGIS add-in wizard toolbar page

7) Sequentially add the three add-in components (combo box, button, tool) from your project into the toolbar and click Finish. You will find that the Config.esriaddinx document is updated to include a new XML element named “Toolbars” which includes four levels of nested elements. The three new add-in components you create are included under the “Items” element (Figure 16).

```
<Toolbars>
  <Toolbar id="NAU_TestAddIn_TestAddIn_Toolbar"
           caption="TestAddIn Toolbar" showInitially="false">
    <Items>
      <ComboBox refID="NAU_TestAddIn_ArcGISOnlineComboBox" />
      <Button refID="NAU_TestAddIn_ZoomToLayerButton" />
      <Tool refID="NAU_TestAddIn_AddGraphicsTool" />
    </Items>
  </Toolbar>
</Toolbars>
```

Figure 16 The Toolbars element in the XML document

In an ArcGIS Desktop add-in project, no VB code is needed to define a toolbar. Since no new file is created for the toolbar, you can find that no new item is added in the Solution Explorer. A toolbar and its contents are simply defined by the “Toolbars” element in the XML document.

To test the toolbar, you can run the program to open ArcMap, then

1) Click the ArcMap menu Customize >> Customize Mode… to open the Customize dialog
2) In the Toolbars tab page, scroll up and down to find your toolbar and check it (Figure 17)

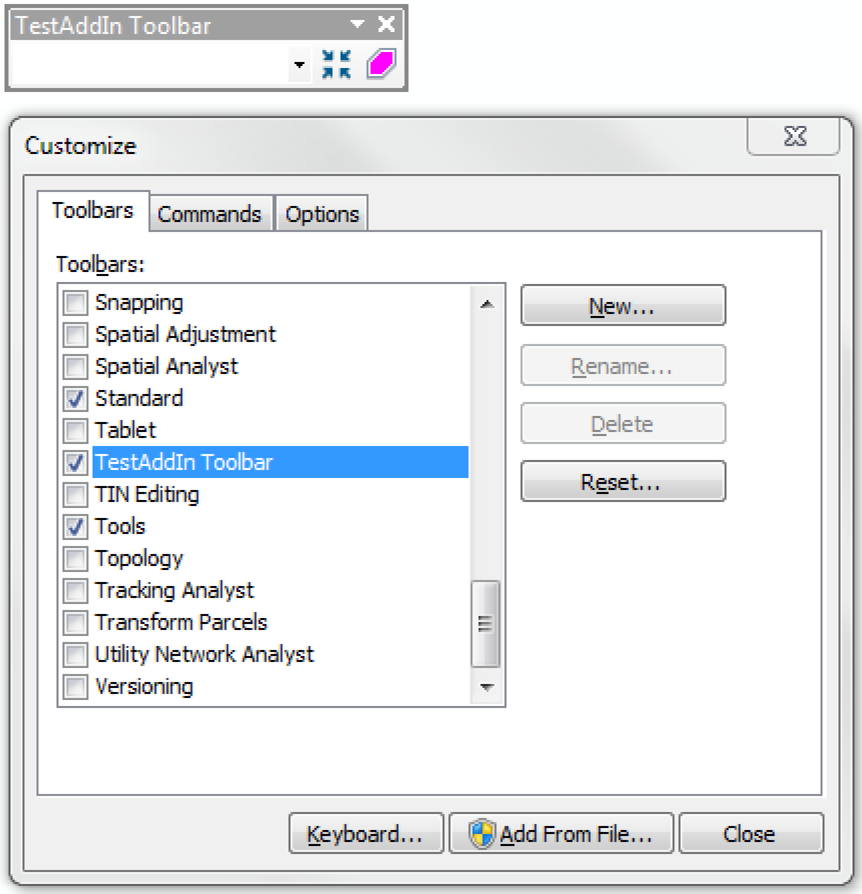

Figure 17 To display the new toolbar

3) Since the three add-in components were previously added into some existing toolbar, you may want to remove them from that toolbar. To do so, you can drag and drop them out of the toolbar and release them in the map area with the Custom window open.

Finally, you may want to know where the compiled binary add-in file is located. It is in folder bin\debug several levels down your solution folder (Figure 18).

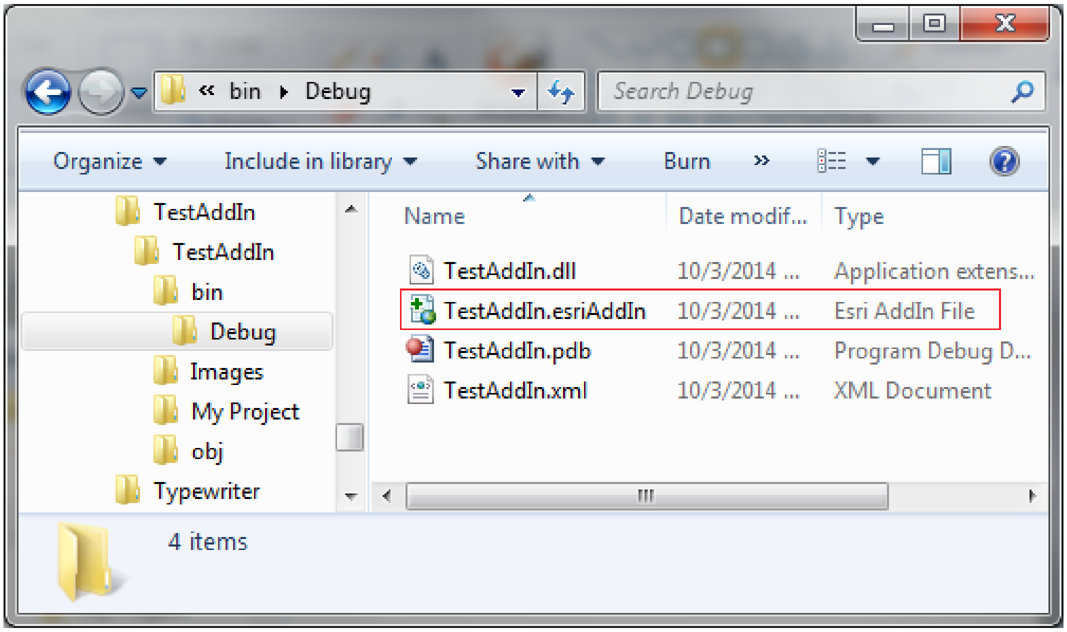

Figure 18 Location of the binary add-in file

When you run the program and open ArcMap, this binary add-in file is created and automatically installed in ArcMap. It does not hurt to leave this add-in installed in ArcMap. However, if you do not need it, you can use the ArcMap Add-In Manager to remove it from ArcMap. To do so, click menu Customize >> Add-In Manager to open the Add-In Manager (Figure 11), then select the add-in and click the “Delete This Add-In” button. If you want to use this add-in later, you can double click on the binary add-in file to install it in ArcMap.

Tips:

1) To avoid repeatedly scrolling left and right when you read the XML configuration document, you can chop long XML lines into multiple lines. A line can be broken at almost any point except inside a keyword.
2) Do you know that you can change images for add-in buttons and tools? The operation is very simple: In Windows Explorer, find the folder named "Images" in your project, copy the new image you want to use into this folder, rename or delete the old image, and then rename the new image to the EXACT NAME of the old image for the button or tool. Finally, run the program and you will find the icon for the button or tool changed.
3) VB.NET provides two message boxes. The traditional one is a function "msgbox" and the newer .NET-based one is a "MessageBox" object. To use msgbox function, you can simply type a quoted message after the function name. The .NET-based message box (MessageBox) is an object that you need to call its "Show" method to display a message. Because the MessageBox object is in namespace Windows.Forms which is not imported by the ArcGIS add-in project template by default, you have to refer to its full namespace in order to use this message box in add-in projects, i.e. Windows.Forms.MessageBox.Show("a message"). To avoid typing the full namespace every time when you use the Messagebox object, you can add a statement "Imports Windows.Forms" on top of the class module. The Show() method of the MessageBox object has a series of parameters that customize the box. When you call this method, watch the IntelliSense for more arguments.

Summary and Highlights

An ArcGIS desktop add-in is a software program that runs in ArcGIS environment such as ArcMap and ArcCatalog. An add-in can contain multiple components such as buttons, tools, combo boxes, and toolbars. A compiled add-in file is a zip compressed file with extension ".esriAddin". You can install an add-in in its targeted ArcGIS application by double clicking this file.

An ArcGIS add-in button is like a Windows button control. When you click on a button, it performs some task. An ArcGIS add-in tool is a toggle button. When you click on it nothing happens except that it is selected. With a tool selected, you can click or drag on the map to carry out intended tasks.

An XML document is a text file that uses user defined tags to encode information in a hierarchical structure. An ArcGIS add-in project uses an XML document named "Config.esriaddInx" to maintain project configurations and add-in component definitions. In this document, the top level element is the ESRI.Configuration element. Inside the top-level element there are a number of metadata elements that describe the add-in, among which the AddInID element maintains a unique ID of the add-in. The Targets element maintains the targeted ArcGIS application and version. The AddIn element includes all add-in components created in the current project. Each add-in component is assigned a unique identifier by the project. In an ArcGIS add-in project, a toolbar is defined in the XML configuration document as an element.

The ArcGIS Snippet Finder is a handy tool that may be used to find code snippet for common tasks. When you use the snippet finder to insert code, the insertion point must be inside a class module, but outside any existing procedure (sub or function). Be careful not to insert code snippets inside existing procedures.

Exercises

1. Answer the following questions.
   1) What is an ArcGIS desktop add-in?
   2) What is the file extension of an ArcGIS add-in?
   3) How do you install an add-in in ArcMap?
   4) How do you find whether an add-in is installed in ArcMap?
   5) How do you remove an add-in from ArcMap?
   6) How do you add an existing command into a toolbar in ArcMap?
   7) What are the differences between an add-in button and an add-in tool?
   8) Where do you write your code to implement functionality of a button?
   9) Where do you write your code to implement functionality of a tool?
   10) What is XML?
   11) What is an XML element?
   12) What is an attribute of an XML element?
   13) What does file Config.esriaddinx in an add-in project do?
   14) What properties does the AddIn element have in the Config.esriaddinx file?
   15) What elements are contained in the Commands element in the Config.esriaddinx?
   16) What properties does a Button element have in the Config.esriaddinx?
   17) What properties does a Tool element have in the Config.esriaddinx?
   18) What is ArcGIS Snippet Finder?
   19) When you use the ArcGIS Snippet Finder, where do you insert code snippets?
   20) How do you create add-in controls (such as buttons and tools) in an existing project?
   21) How do you create a new toolbar in an existing add-in project?
   22) Where is the compiled file of an ArcGIS add-in project located?

2. Suppose you are a GIS specialist at the City of Flagstaff GIS Department, you are recently assigned a task to create an ArcGIS Desktop add-in project that offers a series of tools and functions to automate some map operations. The project should provide a button that turns on and off multiple layers of a map by one click, a combo box containing a number of fixed map scales to allow quickly zooming to a selected scale, a tool to facilitate bus routes design, and a tool to facilitate bus stop design along with other capabilities.

   Start a new ArcMap add-in project in VB and creates a toolbar containing a few add-in controls listed below. Please name your add-in project "CityGIS" and caption the toolbar as "City GIS Toolbar".

Add-in	**Name**	**Image**	**Purpose**
Button	LayersOnOffButton		On click to turn on/off all layers
Tool	AddBusStopTool		For creating bus stops
Tool	AddBusRouteTool		For creating bus routes
Combo box	MapScalesComboBox	(none)	A combo box containing map scales

   Images have been created for the project and are available in the Images folder on the CD-ROM.

   You do not need to write a lot functional code for these add-in controls in this exercise. You only need to write code to add three map scales, 1:5,000, 1:10,000, and 1:25,000, in the add-in combo box. This can be done by adding three lines of code in the constructor (Sub New()) of the combo box class as the following:

```
Public Sub New()
    Me.Add("1:5,000")
    Me.Add("1:10,000")
    Me.Add("1:25,000")
End Sub
```

To test if the LayersOnOffButton add-in control reacts to clicking, please simply add a message box in the OnClick() event procedure of the button.